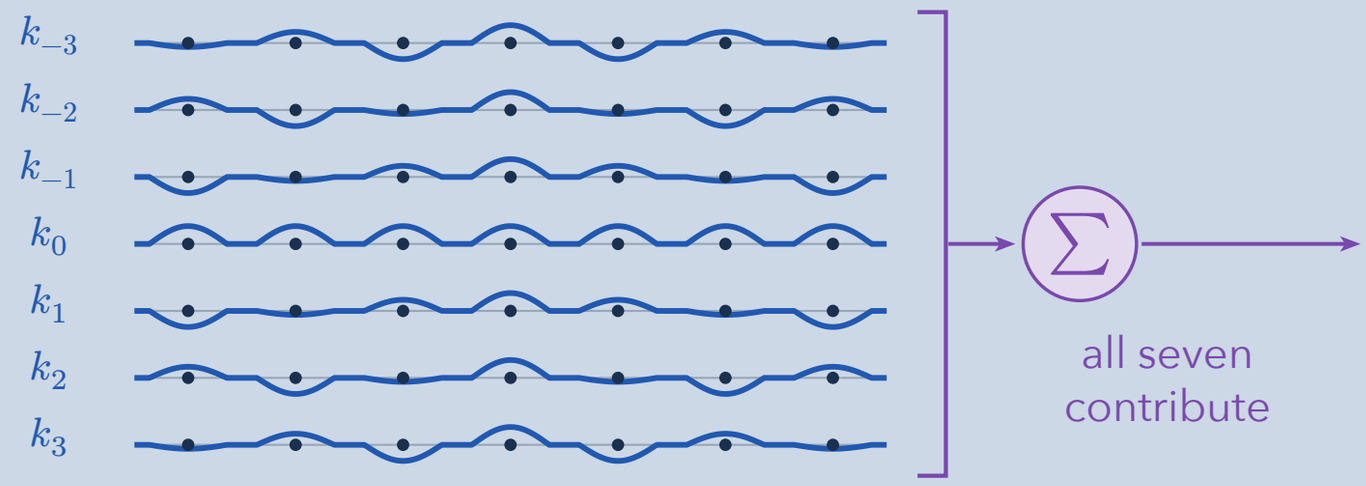
From extended waves to local orbitals and their centers.

● nucleus ◆ Wannier center

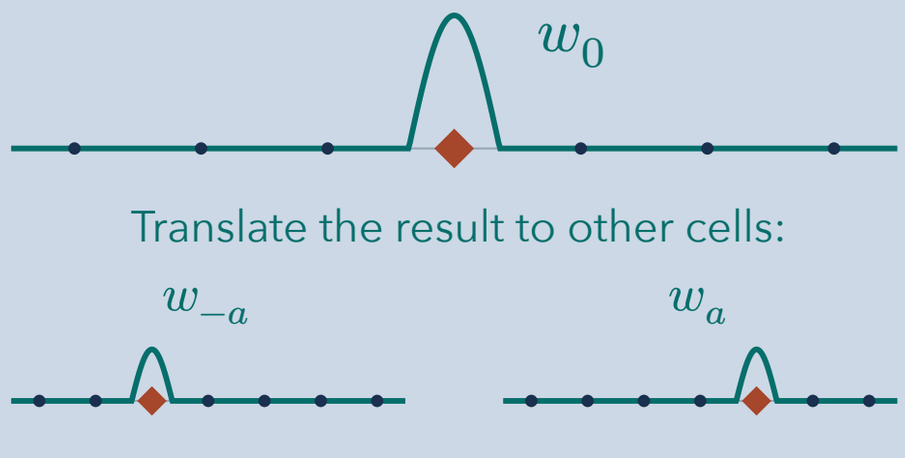
1 CHOOSE PHASES, THEN ADD ALL THE WAVES

One-band toy model • seven states



Real parts shown; $k_j = \frac{2\pi j}{7a}$.

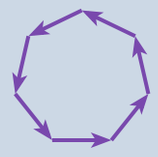
One localized Wannier orbital



At the target cell: phases reinforce

sum = 7

At a neighboring cell: phases cancel



sum = 0

Each arrow is a unit complex phase. Add head to tail; the Fourier prefactor sets normalization.

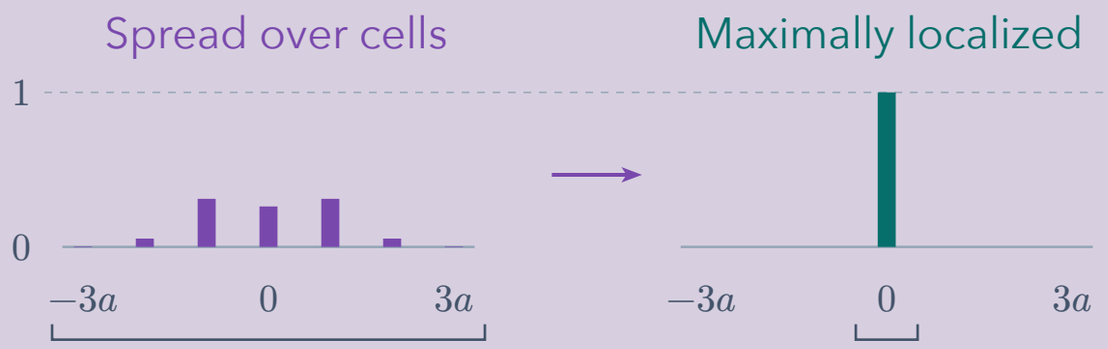
$$w_R(x) = \frac{1}{\sqrt{N_k}} \sum_k e^{-ikR} e^{i\varphi(k)} \psi_k(x)$$

$w_R(x)$: localized Wannier orbital for cell R.
x: position coordinate in real space.
R: lattice position of the chosen cell.
a: lattice spacing; R is a multiple of a.
$\sum_k$: sum over the sampled wavevectors.

$\psi_k(x)$: normalized, extended Bloch state.
k: sampled crystal wavevector.
N_k: number of sampled k points (here 7).
$\varphi(k)$: chosen Bloch phase (here 0).
$\frac{1}{\sqrt{N_k}}$: normalization for orthonormal states.

2 CHOOSE LOCALIZATION

Same physical state; change the basis.



Minimize $\Omega = \sum_n \left(\langle r^2 \rangle_n - |\overline{\boldsymbol{r}}_n|^2 \right)$.

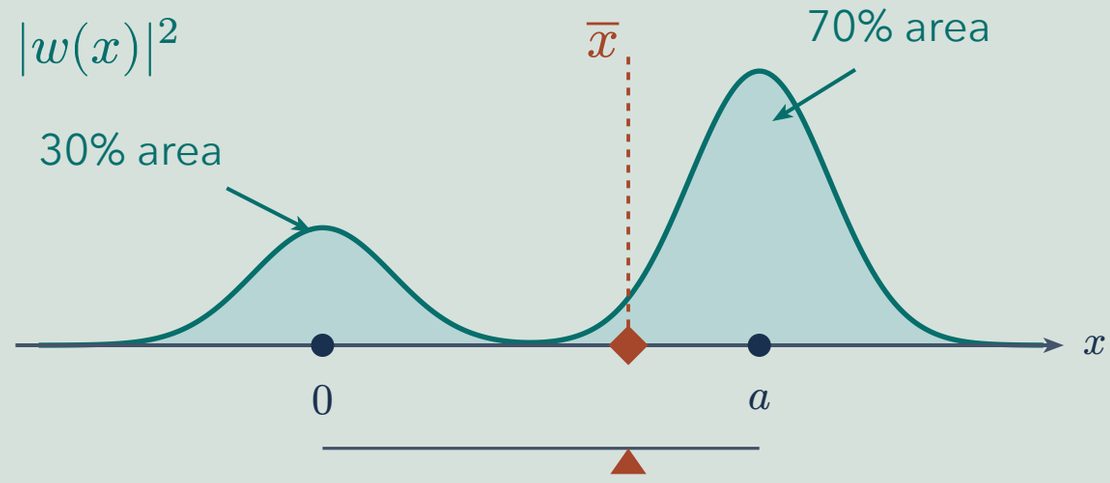
Total occupied density: unchanged



Same fully occupied band subspace.

3 MEASURE THE CENTER

Probability density ⟶ mean position



$$\overline{x} = 0.3(0) + 0.7(a) = 0.7a$$

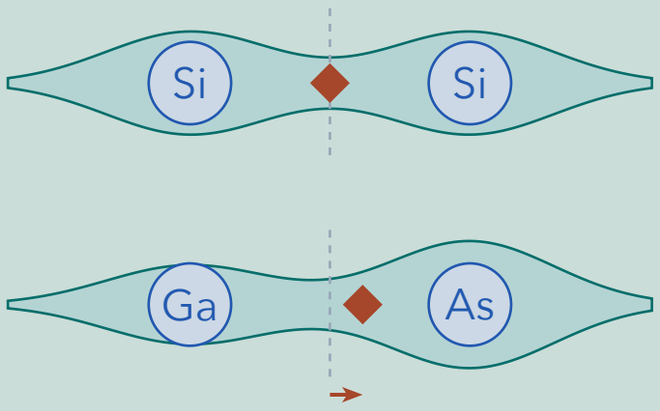
The mean need not coincide with a peak or nucleus.

$$\overline{\boldsymbol{r}}_n = \int \boldsymbol{r} |w_{n\mathbf{0}}(\boldsymbol{r})|^2 \mathrm{d}^3 r$$

4 USE THE LOCAL PICTURE

A READ CHEMICAL BONDS

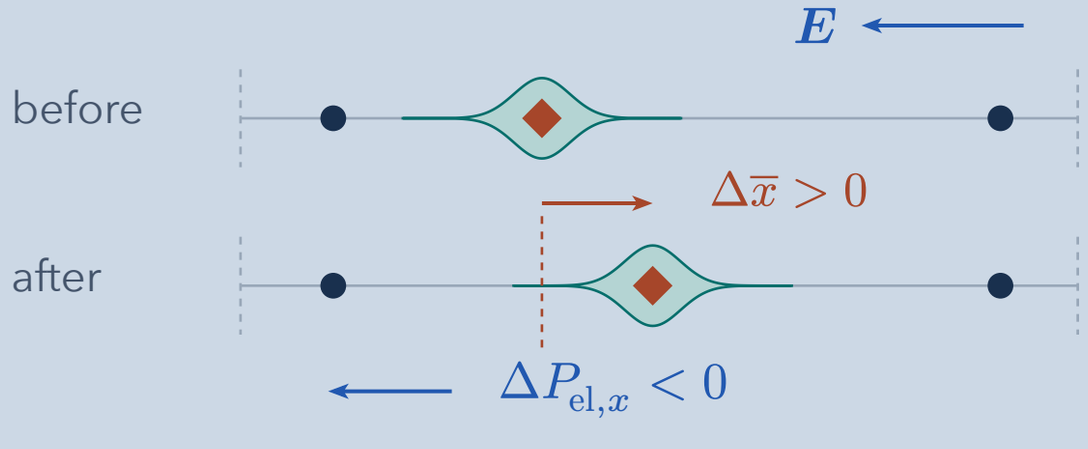
Valence orbitals; density shapes schematic.



Center shifts reveal local bond polarity.

B FOLLOW A PHYSICAL RESPONSE

Apply an electric field; keep ions fixed.



$$\Delta P_{\mathrm{el},x} = -\frac{e}{V_{\mathrm{cell}}} \sum_n f_n \Delta \overline{x}_n$$

$e > 0$; f_n is the occupation (2 for a pair). Fixed cell volume V_{cell}; remain insulating.

Multiband: unitary U

$$w_{n\boldsymbol{R}} = \frac{1}{\sqrt{N_k}} \sum_{\boldsymbol{k}} e^{-i\boldsymbol{k}\cdot\boldsymbol{R}} \sum_m U_{mn}(\boldsymbol{k}) \psi_{m\boldsymbol{k}}$$

Individual centers depend on gauge U. Track them across cell boundaries; add ions for total polarization.